

GRAP-R method to consider the nearby rank-deficient points where apocalypse often occurs. Line search along approximate projections contributes to an Armijo condition. Both steps protect the sequences generated by GRAP-R from converging to non-stationary points, which renders GRAP-R apocalypse-free.

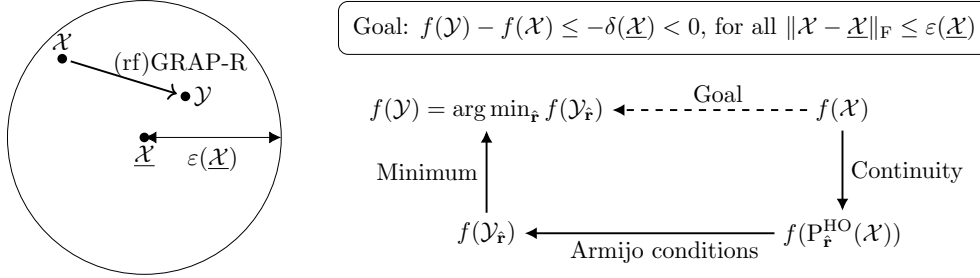


FIG. 2. Definition and validation of the sufficient decrease condition in convergence analysis. In a neighborhood of a non-stationary point $\underline{X}$, the function value decreases by more than $\delta(\underline{X})$ from $\mathcal{X}$ to $\mathcal{Y}$. The roadmap illustrates how the (rf)GRAP-R method achieves the sufficient decrease condition.

Alternatively, we consider retraction-free search directions that line searches never leave $\mathcal{M}_{\leq \mathbf{r}}$. In a similar manner, we adopt the framework in Figure 1 and propose a new retraction-free GRAP-R method (rfGRAP-R). In contrast with the GRAP-R method where all singular values of unfolding matrices are required, the rfGRAP-R method only detects the smallest singular value of unfolding matrices, which is in favor of saving computational costs. We prove that the rfGRAP-R method also converges to stationary points by verifying a sufficient decrease condition in Figure 2.

In addition, numerical results validate that the proposed methods indeed converge to stationary points, and the rfGRAP-R method performs better than the GRAP-R method due to the simpler rank-decreasing mechanism.

Organization. We introduce the geometry of Tucker tensor varieties and notation for tensor operations in section 2. In section 3, we provide an explicit representation of the normal cone of Tucker tensor varieties, and propose approximate projections. Based on two different projections, we propose two first-order methods and prove that all accumulation points of sequences generated by these methods are stationary in sections 4 and 5. Section 6 reports the numerical performance of proposed methods in tensor completion. Finally, we draw the conclusion in section 7.

2. Geometry of Tucker tensor varieties. In this section, we introduce the preliminaries of variational analysis for low-rank matrices. Then, we describe the notation in tensor operations and Tucker tensor varieties.

Given a nonempty subset C of $\mathbb{R}^{n_1 \times n_2 \times \cdots \times n_d}$, the (Bouligand) *tangent cone* of C at $\mathcal{X} \in C$ is defined by $T_{\mathcal{X}}C := \{\mathcal{V} \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_d} : \exists t^{(i)} \rightarrow 0, \mathcal{X}^{(i)} \rightarrow \mathcal{X} \text{ in } C, \text{ s.t. } (\mathcal{X}^{(i)} - \mathcal{X})/t^{(i)} \rightarrow \mathcal{V}\}$. The set $N_{\mathcal{X}}C = (T_{\mathcal{X}}C)^{\circ} := \{\mathcal{W} \in \mathbb{R}^{n_1 \times n_2 \times \cdots \times n_d} : \langle \mathcal{W}, \mathcal{V} \rangle \leq 0 \text{ for all } \mathcal{V} \in T_{\mathcal{X}}C\}$ is called the *normal cone* of C at $\mathcal{X}$. Note that if C is a manifold, the tangent cone $T_{\mathcal{X}}C$ (normal cone $N_{\mathcal{X}}C$) is a linear space and referred to as the tangent space (normal space). Given a continuously differentiable function f on C , a point $x \in C$ is stationary of f if $\|P_{T_x C}(-\nabla f(x))\|_F = 0$, or $-\nabla f(x) \in N_x C$ equivalently.